MURATOV ARTYOM

2-year MIPT student

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EXPERIENCE

Summer Internship at Huawei

Position: Assistant Engineer

Jun 2025 – Aug 2025

- Developed an optimization pass for an ARM binary optimizer using an LLVM-based internal framework.
- The pass replaces simple comparison patterns with CSEL instructions, reducing branch mispredictions and improving code density.
- Collaborated with senior engineers during code reviews to integrate the pass into the main optimizer pipeline.

EDUCATION

Introduction to industrial programming in C++ course, 2025 – present

MIPT course, Lecturer: Konstantin Vladimirov

Introduction to tensor compilers course, 2026 – present

MIPT and Sber course, Lecturers: Alexey Eganyan, Vladislav Loznovenko

System programming and compiler technology, course 2024 – 2025

MIPT course, Lecturer: Ilya Dedinsky

Bachelor of Radio Engineering and Computer Technology, 2024 – present

Moscow Institute of Physics and Technology

MAIN PROJECTS

Implementation of bitonic sort algorithm with GPU (C++, OpenCL), 2026

- Developed an OpenCL-based GPU sorting algorithm achieving up to 8x speedup on 1M elements over std::sort on CPU (tested on Intel(R) UHD Graphics).
- Implemented RAII wrappers for OpenCL objects to ensure exception safety and resource management.
- Optimized kernel by using local memory and workgroup-level parallelism.

Interpreter for a custom programming language (C++, Flex, Bison), 2026

- Built a full-featured interpreter with lexical analysis (Flex), parsing (Bison), and AST-based execution.
- Implemented scoping, error reporting with line/column info, and a visitor pattern for AST traversal.
- Successfully runs test programs with variables, loops, and conditionals.

Tensor Compiler (C++, Protobuf), 2026

- Designed a computational graph data structure for neural networks. Implemented a parser to load .onnx models and convert them into the internal graph representation.
- Added support for 6+ neural network operations (Add, Mul, Conv, Relu, MatMul, Gemm) with full attribute parsing.

Solving the triangle intersection problem in 3D (C++, OpenGL), 2025

- Engineered a geometric algorithm for triangle intersection in 3D space using orientation predicate.
- Implemented an OpenGL-based application with custom shaders to visualize triangles and their intersections.

Compiler for a custom programming language (C/C++, x86-64 ASM), 2025

- Developed a multi-pass compiler: recursive-descent frontend, IR with optimizations (constant folding, neutral expression removal), and backend targeting x86-64 and a simulated custom CPU.
- Generated assembly code that passes all provided test cases.

High-performance word search engine with Hash Table (C/C++, x86-64 ASM, Intrinsics), 2025

- Developed a hash table search engine processing 75M queries on 20K unique words, achieving 96% speedup (14.36s → 7.33s) against -O3 baseline.
- Optimized hot paths using SIMD intrinsics (__m128i) for string comparison and hardware CRC32 instructions for hashing, guided by Perf profiling.

TECHNICAL SKILLS

Programming Languages

C++ (modern), C, NASM (x86-64), Python, OpenCL, GLSL

Tools & Technologies

Git, CMake, Make, Bash, Radare2, Perf, Godbolt, Flex, Bison, OpenGL

Languages

Russian (Native), English (B2)

INTERESTS

- Member of the MIPT table tennis team.
- Reading classic literature, business and self-development books.